Junwoon Lee

Website: https://junwoonlee.github.io
Email: junwoon@umich.edu

RESEARCH INTEREST

Spatial Intelligence, Neural Representation, Multi-Agent Systems, Field Robotics

EDUCATION

University of Michigan *Aug. 2025 –*
Ph.D. Candidate in Robotics
- Advisor: Prof. Yulun Tian

University of Tokyo *Apr. 2023 – Mar. 2025*
M.E.S. in Human & Engineered Environmental Studies
- Advisor: Prof. Atsushi Yamashita
- Thesis: Switching-based Multi-modal SLAM for Extreme and Degraded Environments

University of Osaka *Apr. 2017 – Mar. 2023*
B.E. in Mechanical Engineering (Military Service included)
- Advisor: Prof. Masamitsu Kurisu
- Thesis: LiDAR-visual SLAM for Online Mapping of Unpaved Road Surface

PUBLICATIONS

[1] **LatentAM: Real-Time, Large-Scale Latent Gaussian Attention Mapping via Online Dictionary Learning**
Junwoon Lee, Yulun Tian
IEEE Robotics and Automation Letters (RA-L), 2026. [Link]

[2] **CubeDVO: Cubemap-Spherical Deep Visual Odometry for a Monocular 360-Degree Camera**
Taisei Ando, Junwoon Lee, Takuya Igaue, Kohtaro Nakamura, Mitsuru Shinozaki, Toshihiro Kitajima, Qi An, Atsushi Yamashita
IEEE Robotics and Automation Letters (RA-L), 2026. [Link]

[3] **Robot Localization by Data Integration of Multiple Thermal Cameras in Low-light Environment**
Masaki Chino, Junwoon Lee, Qi An, Atsushi Yamashita
International Journal of Automation Technology, 2025. [Link]

[4] **Efficient Distortion Mitigation in Equirectangular Images for Two-View Pose Estimation**
Taisei Ando, Junwoon Lee, Mitsuru Shinozaki, Toshihiro Kitajima, Qi An, Atsushi Yamashita
International Journal of Automation Technology, 2025. [Link]

[5] **Self-TIO: Thermal-Inertial Odometry via Self-supervised 16-bit Feature Extractor and Tracker**
Junwoon Lee, Taisei Ando, Mitsuru Shinozaki, Toshihiro Kitajima, Qi An, Atsushi Yamashita
IEEE Robotics and Automation Letters (RA-L), 2025. Presented at IROS'25. [Link]

[6] **TC-LTIO: Tightly-coupled LiDAR Thermal Inertial Odometry for LiDAR and Visual Odometry Degraded Environments**
Junwoon Lee, Taisei Ando, Mitsuru Shinozaki, Toshihiro Kitajima, Qi An, Atsushi Yamashita
International Conference on Control, Automation and Systems (ICCAS), 2024. [Link] (Best Paper Award)

[7] **Highly Accurate and Fast Two-view Pose Estimation by Fast Reduction of Spherical Image Distortion Effects**
Taisei Ando, Junwoon Lee, Mitsuru Shinozaki, Toshihiro Kitajima, Qi An, Atsushi Yamashita
International Conference on Control, Automation and Systems (ICCAS), 2024. [Link]

[8] **Switch-SLAM: Switching-Based LiDAR-Inertial-Visual SLAM for Degenerate Environments**
Junwoon Lee, Ren Komatsu, Mitsuru Shinozaki, Toshihiro Kitajima, Hajime Asama, Qi An, Atsushi Yamashita
IEEE Robotics and Automation Letters (RA-L), 2024. Presented at ICRA@40. [Link]

[9] **Three-dimensionalized Feature-Based LiDAR-visual Odometry for Online Mapping of Unpaved Road Surface**
Junwoon Lee, Masamitsu Kurisu, Kazuya Kuriyama
Journal of Field Robotics, 2024. [Link]

RESEARCH EXPERIENCE

Research Assistant, University of Michigan *Aug. 2025 –*
Scalable Spatial Intelligence Lab.

- Developing a VLM features-embeded Gaussian Splatting SLAM system.
- Developing a LiDAR-visual-GNSS-inertial Multi-robot experimental platform.

Research Assistant, University of Tokyo *Apr. 2023 – Jul. 2025*
Real World Robot Informatics Lab.

- Developed a multi-modal SLAM system for complex scenes.
- Developed a self-supervised point tracker for thermal inertial odometry.

Research Assistant, University of Osaka *Apr. 2022 – Mar. 2023*
Komatsu MIRAI Construction Equipment Cooperative Research Center

- Suggested an intensity-weighted point cloud registration.
- Developed a mapping system for an unpaved road surface.

HONORS AND AWARDS

Department Fellowship *Sep. 2025 – Aug. 2026*

- Fellowship for 1st-year doctorate studies.

BOOST NAIS AI Fellowship (declined) *Apr. 2025 – Mar. 2028*

- Full Fellowship (Ph.D.) at UTokyo, $75,000+ USD

Dean's Award *Mar. 2025*

- Graduated at the top of the department (1/39)

Best Paper Award *Oct. 2024*

- ICCAS'24 (1/400)

IEEE RAS Travel Grant *Sep. 2024*

- ICRA@40, $2,000 USD

Rotary Yoneyama Memorial Foundation Scholarship *Apr. 2023 – Mar. 2025*

- Full scholarship (M.S.), $30,000+ USD

Korea-Japan Joint Government Scholarship *Apr. 2017 – Mar. 2023*

- Full scholarship (B.S.), $75,000+ USD

TEACHING

Teaching Assistant, UTokyo FEN-SC3102S1 Exercises for Mathematics 2C *Apr. 2024 – Jul. 2024*

MENTORING

Siddhant Bohra (B.S., University of Michigan), Full stack mobile robot-system implementation *May. 2026 –*
Arnav Shah (M.S., University of Michigan), Foundation model-based loop closure *Feb. 2026 –*
Yiyu Wang (B.S., University of Michigan), Sensor suite design, SLAM benchmarking *Sep. 2025 –*
Kohei Yamaguchi (B.S., University of Tokyo), Learning-based point cloud registration *Apr. 2025 –*
Taisei Ando (Ph.D./M.S./B.S., University of Tokyo), Learning-based omnidirectional SLAM *Apr. 2023 –*

SERVICES

Academic Reviewer

- T-RO (2025), RA-L (2024–2026), ICRA (2025–2026), IROS (2025–2026), T-Mech (2026), T-ASE (2024–2025), AURO (2026), Journal of Field Robotics (2025), IEEE Sensors Journal (2024–2025), ICCAS (2024), Scientific Reports (2024)

Sergeant, Republic of Korea Army *Apr. 2020 – Oct. 2021*

- Served as a frontline guardian at a coastline observation post in the 23rd Security Brigade

PATENT

1. Kaoru Adachi, Masamitsu Kurisu, Junwoon Lee, "Terrain Detection System and Method," *Japanese Patent app:2023-105215 / open:2025-005158*, Filed on June 27, 2023.

REFERENCES

References available upon request.